

Professional Summary

Senior AI/ML Engineer and Data Scientist with 5+ years of hands-on experience in Python programming, Big Data technologies, cloud platforms (GCP, Azure), and enterprise-scale machine learning solutions. Demonstrated expertise in **Generative AI, LLMs, and agentic frameworks** (LangChain, RAG pipelines, vector databases), with deep proficiency in **MLOps, data engineering, and API development**. Proven track record of architecting and deploying production-grade AI/ML systems using **BigQuery, PySpark, Vertex AI, and serverless architectures**, driving measurable business impact through advanced analytics, time-series forecasting, and intelligent automation.

Core Competencies

Programming & Frameworks: Python (Expert: FastAPI, Flask, Django, pandas, scikit-learn, TensorFlow, PyTorch, Hugging Face), Java (Spring Boot), JavaScript/TypeScript (React, Next.js), SQL, Swift

Big Data & Cloud: BigQuery, PySpark, Hadoop, Airflow • GCP (Vertex AI, Cloud Run, Dataproc), Azure (OpenAI, Cognitive Services), AWS (S3, EC2, SageMaker) • Terraform, Docker, Kubernetes

AI/ML & GenAI: ML Models (XGBoost, time-series forecasting), Deep Learning (CNNs, LSTMs, Transformers) • LLMs (GPT-4, Gemini, Claude), LangChain, RAG pipelines, Vector DBs (Pinecone, Chroma), Prompt Engineering

MLOps & DevOps: CI/CD (Tekton, Airflow, GitHub Actions), Model Deployment, Monitoring • RESTful APIs, WebSocket, OAuth2, JWT • Nginx, Unicorn, Load Balancing

Professional Experience

Ford Motor Company

Detroit, MI, USA

Senior Data Scientist & AI/ML Engineer

Aug 2023 – Present

- Architected **Python applications** (FastAPI, Flask) processing **100K+ data points/sec** with OOP, advanced data structures, and RESTful APIs for EV analytics microservices
- Built **Big Data pipelines** using **PySpark and BigQuery**, processing **10+ TB** of EV charging data for real-time analytics and predictive modeling
- Developed **Generative AI agent** with **LangChain and LLMs** for automated code complexity analysis and GCP cost optimization, reducing manual review time by **60%**
- Deployed **MLOps infrastructure on GCP** (Cloud Run, Vertex AI, Dataproc) with **CI/CD pipelines** (Tekton, Airflow), achieving **99.9%** uptime
- Built ML models (XGBoost, Transformers: TFT, Autoformer) with Optuna optimization, improving predictions by **10%** and reducing training time by **40%**
- Implemented **Terraform IaC** for multi-environment GCP deployments with IAP authentication and secure API Gateway configurations

CereVu Medical, Inc.

San Francisco Bay Area, CA

Software Engineer & AI/ML Researcher

Jun 2023 – Sep 2024

- Developed **Python microservices** (Flask, FastAPI) with RESTful APIs and WebSocket for real-time biosignal streaming, supporting **10K+ concurrent connections**
- Designed **MLOps CI/CD pipelines** (Tekton, AWS CodeDeploy) with Docker, achieving **30%** faster releases and zero-downtime deployments
- Built ETL pipelines with MongoDB and AWS (S3, EC2) for biosignal data processing and real-time ML analytics
- Deployed deep learning models (LSTM, CNN) improving patient monitoring accuracy by **15%**; architected SwiftUI iOS app with Firebase integration

Texas A&M University

Research Assistant & ML Engineer

College Station, TX

Sep 2021 – May 2024

- Built **Python ML pipelines** (TensorFlow, PyTorch) for wearable stress detection with **76%** accuracy improvement via deep learning (LSTM, CNN) and sensor fusion
- Developed iOS app (*GlucoseCoach*) in SwiftUI with Flask backend for real-time ML inference and personalized health monitoring
- Created **computer vision models** for COVID-19 compliance (**98%** accuracy) and gait analysis (**93%** accuracy using Conv-LSTM)

Selected Projects

Clink - Social Platform: [moeinrazavi/Clink](#) • Full-stack app (React Native, TypeScript, Remix) with Prisma ORM, PostgreSQL, OAuth2/JWT • Real-time chat (WebSocket), Plaid payment integration, Apple MapKit

AI Agents Hub: [moeinrazavi/AI-Companion](#) • Spring Boot 3 + Spring AI + LangChain • RAG pipeline with Pinecone vector DB • GCP (Speech-to-Text, TTS, Firestore) + Azure Cognitive Services

AI Companion ML: [moeinrazavi/AI-companion-ML](#) • Python Flask + LangChain + Azure OpenAI • Emotion detection (Hugging Face Transformers) with 90%+ accuracy • Real-time NLP pipeline

Stock Market Prediction: [moeinrazavi/StockMarketPrediction](#) • Python LLM ensemble (GPT-4) trading system • Streamlit dashboard with automated backtesting • Sharpe ratio & risk analysis

Healthcare Dashboard: [moeinrazavi/cerevuReactDashboard](#) • Next.js 14 + TypeScript + Material-UI • JWT auth, MongoDB, Redux • Real-time biosignal data visualization (Recharts, Chart.js)

Education

Texas A&M University

Ph.D. in Industrial Engineering (Data Science)

GPA: 4.0/4.0

College Station, TX

2021 – 2023

Texas A&M University

M.S. in Computer Science

GPA: 4.0/4.0

College Station, TX

2019 – 2021

Certifications

Advanced MLOps Specialization (Coursera) • **NLP Specialization** (Coursera) • **Deep Learning Specialization** (DeepLearning.AI) • **Generative AI with LLMs** (AWS)

Selected Publications

Razavi, M., et al. (2024). *Integrated Eye-Tracking and EEG Data for VR Assessments*. IHSI • (2023). *ML & Deep Learning for Stress Detection: A Scoping Review*. JMIR • (2022). *Automatic Physical Distancing & Mask Monitoring*. SN Computer Science • (2021). *OpenSync: Synchronizing Measures in Neuroscience*. J. Neuroscience Methods • Co-author on IEEE HPCA & iScience publications